

svgdigitizer: A Python package for recovering data from plots of scientific publications

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Summary

Scientific knowledge is communicated primarily through publications, where quantitative data is most often presented as two-dimensional figures rather than as machine-readable tables. Once a study is published, the underlying numerical data frequently becomes inaccessible — particularly for older literature where digital data was never archived. `svgdigitizer` is a Python library and command-line tool that recovers the data from such figures by digitizing curves manually traced in carefully annotated Scalable Vector Graphics (SVG) files, a XML-based format that allows programmatic access to its elements. The user imports a figure image into a vector graphics editor, e.g., Inkscape (Inkscape Project, 2025), traces the curve of interest with Bézier paths, and annotates the coordinate system with text labels (Figure 1 left). `svgdigitizer` then applies an affine coordinate transformation to map SVG pixel coordinates to plot data coordinates and samples the Bézier paths at user-defined intervals to produce a tabular dataset. Physical unit information encoded in axis labels is parsed via `astropy.units` (Collaboration et al., 2022) and propagated to the output. The digitized data is exported as a *frictionless datapackage* (Frictionless Data, 2026) — a CSV file accompanied by a JSON descriptor containing units, metadata, and bibliographic information, enabling FAIR (Findable, Accessible, Interoperable, and Reusable)-compliant (Wilkinson et al., 2016) data (Figure 1 right).

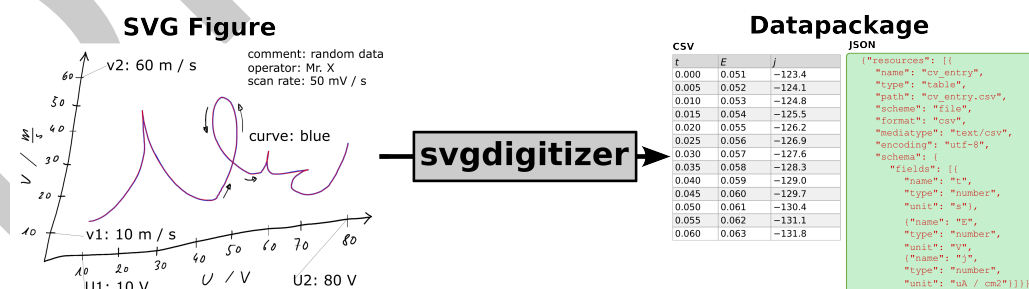


Figure 1: `svgdigitizer` transforms an annotated SVG figure (left) into a frictionless datapackage consisting of a CSV file with the curve data and a JSON descriptor containing axis units and extracted metadata (right).

Statement of Need

The ability to extract quantitative data from published figures is essential across scientific research. Meta-analyses, literature benchmarking, and database curation all require numerical

29 data that is often available only in graphical form. This problem is particularly acute for
30 historical literature, where measurements were never stored digitally beyond the printed figure,
31 but it equally affects modern publications that do not accompany figures with machine-readable
32 raw data. While in most cases specific values or summary statistics of measurement data is
33 sufficient, and which can often be extracted from the text body manually or large language
34 models, in some cases the entire curve must be preserved.

35 `svgdigitizer` addresses this need for researchers who require systematic, reproducible, and
36 metadata-rich digitization of the entire curve in a scientific plot (see [Figure 1](#)). Beyond one-off
37 data extraction, it is designed for workflows in which the same figure may be re-digitized with
38 different sampling parameters, or in which digitized data from many publications must be
39 aggregated into a common curated dataset. The FAIR frictionless datapackage output format
40 enables digitized data to be shared, versioned, and integrated into databases with minimal
41 friction.

42 A primary motivation for `svgdigitizer` is the community database of cyclic voltammograms
43 ([Albert K. Engstfeld et al., 2026](#); [The echemdb development team, 2021](#)), where contributions
44 are curated as SVG files digitized with `svgdigitizer`. This workflow enables electrochemists
45 to contribute historical literature data systematically, with each entry carrying full provenance
46 metadata and physically meaningful units.

47 State of the Field

48 Several tools are available for extracting numerical data from published plots, ranging from
49 web-based interfaces to windows-only desktop and cross-platform applications, and spanning
50 both open-source and commercial licenses. Examples include WebPlotDigitizer ([Rohatgi, 2024](#)),
51 starry-digitizer (built on WebPlotDigitizer) ([Katsura et al., 2025](#)), Engauge Digitizer ([Mitchell
52 et al., 2020](#)), or OriginPro ([OriginLab Corporation, 2026](#)). These tools typically provide
53 interactive workflows for recovering data points from graphical representations through manual
54 selection or semi-automated detection methods, enabling plotted information to be converted
55 into reusable numerical datasets. Furthermore they are well-suited to one-off, GUI-driven data
56 extraction. However, they usually do not allow exporting physical units, nor produce structured
57 metadata, nor offer a programmable Python API for integration into automated workflows.

58 In the following, we briefly summarize the current limitations of the available tools.
59 `svgdigitizer` was designed as a distinct tool rather than a contribution to existing projects for
60 several reasons. First, the commonly available tools work with raster images where individual
61 data points are selected on the curve by mouse clicks and to some extent by auto detection
62 methods, which then present the final dataset. `starry-digitizer` allows interpolating between
63 selected datapoints to get a more fine grained dataset. Second, common tools sample across
64 one of the axis (usually the x -axis). This presents an issue when a curve follows a path that
65 has multiple y -axis values per x -axis value, leading to scattering of the output data upon
66 sampling along the x -axis. Third, in most cases once the digitization process is finished, the
67 curve-tracing and plot annotation (input data for the extraction process) is not preserved in
68 a reusable, version-controllable artifact that can be reprocessed at any time with different
69 parameters, without repeating the manual work. Fourth, extraction of physical units and the
70 creation of standardized file formats including metadata is a central aspect to render output
71 data reusable, which has only been adopted recently in the `starry-digitizer`. Finally, open
72 source APIs are lacking, which limit the integration with the other scientific ecosystems, such
73 as Python, and the development of domain specific modules to extract specific types of data
74 according to pre defined standards.

Software Design

The `svgdigitizer` module is organized in a layered class hierarchy, each layer adding abstraction over the one below:

- `SVG`: parses the carefully annotated SVG document object model (DOM) and exposes labeled paths and text elements.
- `SVGPlot`: reads axis reference-point labels, e.g., `x1: -40`, `x2: 40`, and constructs an affine transformation from SVG pixel space to plot data space.
- `SVGFigure`: adds unit handling, metadata extraction, and frictionless datapackage output.
- `CV` (`svgdigitizer.electrochemistry`): a domain-specific subclass for cyclic voltammograms.

Two coordinate transformation modes are supported: *axis-aligned* (default, for standard upright plots) and *mark-aligned* (for skewed or scanned figures with non-orthogonal axes).

The carefully annotated SVG files used by `svgdigitizer` contain the following elements, required for the data recovery workflow. A curve which consists of individual points manually placed by the user on the curve, which are connected by Bézier path segments. During the recovery process, these curves are sampled by solving polynomial root-finding problems via `scipy` (Virtanen et al., 2020), yielding data points at configurable equidistant intervals along the x (or y) axis. This enables dense, uniform sampling of curved regions without user-specified point placement. Axis labels contain an identifier (two for each axis), the value, a unit, and in some special cases also additional information, such as `E2: 1 V vs. RHE`. The units follow the `astropy.units` notation, allowing for example, transformation of the data into SI units. The label is decomposed using regular expressions, where the values allow for reconstructing the curve. A text label `scan rate: 50 mV / s` can be added to the curve, which allows reconstructing the time axis of the time series data. Metadata from annotations embedded in the SVG, such as additional text fields with comments or other information about the figure, is extracted as Python dictionary.

Advanced annotation features include support for **scale bars** (deriving axis scale from a labeled reference bar), numeric **scaling factors** (rescaling of an axis often found for a curve in multi-signal overlays), and **scatter plots** (extracting individual data points without Bézier interpolation).

The standard output of `svgdigitizer` are frictionless datapackages, which consist of a CSV (including the curve data) and a JSON (providing information on the axis units and additional metadata from the SVG) file. During the creation of the datapackage, additional metadata can be added from external YAML and BibTeX files into the datapackage descriptor, producing a self-describing output bundle.

A CLI built with `click` (Pallets Projects, 2026) exposes the full feature set without Python knowledge in a convenient way. Notably, the `paginate` command converts PDF pages to SVG and PNG files ready for annotation (where the PNG is linked into the SVG), while optionally renaming (`--rename`) the file base to a sluggified identifier generated from the citation (retrieved with the extracted DOI from the PDF via doi.org) streamlining provenance metadata collection. After curve tracing the `digitize` command yields a datapackage.

Research Impact Statement

`svgdigitizer` underpins the cyclic voltammogram database (Albert K. Engstfeld et al., 2026; The echemdb development team, 2021), a community-curated collection of electrochemical measurements from the literature. Some of the data were extracted for specific scientific purposes to address specific questions in the field of electrochemistry (Bergmann et al., 2023; Heubach et al., 2025). The input data of the database is stored in the above described file formats (SVG and YAML), which allows adjusting curve tracings or resolve other annotation errors by the community. (The echemdb development team, 2021) The output data is made

accessible to the community via the unitpackage (A. Engstfeld et al., 2026) API. Both tools form the data layer of a lightweight file-system-based approach to research data management described in (Albert K. Engstfeld et al., 2025). svgdigitizer was also used by Fritsche et al. to extract geometric features from weld cross sections of refill friction stir spot welds joining additive-manufactured to wrought aluminium alloys (Fritsche et al., 2026).

The package is distributed via PyPI and conda-forge, is archived on Zenodo (Albert Kilian Engstfeld et al., 2026), and has been under active development since June 2021 with over 1,600 commits from multiple contributors.

AI Usage Disclosure

No generative AI tools were used in the development of the svgdigitizer software package and its repository, including code, tests, and documentation. The initial draft of this manuscript was produced with the assistance of Claude Sonnet 4.6 (Anthropic) as a writing aid, based on the documentation of the svgdigitizer project. All content was reviewed, revised, and validated by the authors, who take full responsibility for the accuracy of the manuscript.

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References

- Bergmann, N., Hörmann, N. G., & Reuter, K. (2023). Ab Initio-Based Modeling of Thermodynamic Cyclic Voltammograms: A Benchmark Study on Ag(100) in Bromide Solutions. *Journal of Chemical Theory and Computation*, 19(23), 8815–8825. <https://doi.org/10.1021/acs.jctc.3c00957>
- Collaboration, T. A., Price-Whelan, A. M., Lim, P. L., Earl, N., Starkman, N., Bradley, L., Shupe, D. L., Patil, A. A., Corrales, L., Brasseur, C. E., Nöthe, M., Donath, A., Tollerud, E., Morris, B. M., Ginsburg, A., Vaher, E., Weaver, B. A., Tocknell, J., Jamieson, W., ... Zonca, A. (2022). The astropy project: Sustaining and growing a community-oriented open-source project and the latest major release (v5.0) of the core package*. *The Astrophysical Journal*, 935(2), 167. <https://doi.org/10.3847/1538-4357/ac7c74>
- Engstfeld, Albert K., Hermann, J. M., Hörmann, N. G., & RÜth, J. (2025). A lightweight file system based approach to getting data ready for data management solutions. *Data Science Journal*, 24. <https://doi.org/10.5334/dsj-2025-013>
- Engstfeld, Albert Kilian, Hermann, J. M., Hörmann, N., & RÜth, J. (2026). *Svgdigitizer* (Version 0.14.4). Zenodo. <https://doi.org/10.5281/zenodo.20467078>
- Engstfeld, Albert K., RÜth, J., Hermann, J. M., & Hörmann, N. G. (2026). *Echemdb — a database of electrochemical data*. <https://www.echemdb.org>
- Engstfeld, A., RÜth, J., Hermann, J., & nicohoermann. (2026). *Echemdb/unitpackage: 0.13.1* (Version 0.13.1). Zenodo. <https://doi.org/10.5281/zenodo.19051513>
- Frictionless Data. (2026). *Frictionless data specifications*. <https://specs.frictionlessdata.io>
- Fritsche, S., Draper, J., Batistão, B. F., Toumpis, A., Galloway, A., & Amancio-Filho, S. T. (2026). Microstructure, hook formation and mechanical performance in refill friction stir spot welding of additive manufactured to wrought aluminium joints. *Materials & Design*, 266, 116061. <https://doi.org/10.1016/j.matdes.2026.116061>

- 165 Heubach, M.-K., Schuett, F. M., Mayer, J., Elkhafif, O. W., Engstfeld, A. K., & Jacob,
166 T. (2025). Reproducibility of Electrochemical Measurements with Ionic Liquids: Role of
167 Supplied Batches, Water and Adventitious Oxygen. *Topics in Catalysis*, 68(16), 1910–1923.
168 <https://doi.org/10.1007/s11244-025-02123-x>
- 169 Inkscape Project. (2025). *Inkscape*. <https://inkscape.org>
- 170 Katsura, Y., Kumagai, M., Mato, T., Takada, Y., Ando, Y., Fujita, E., Hosono, F., Koyama,
171 E., Mudasar, F., Phuong, T. N. T., & others. (2025). Starrydata: From published plots to
172 shared materials data. *Science and Technology of Advanced Materials: Methods*, 5(1),
173 2506976. <https://doi.org/10.1080/27660400.2025.2506976>
- 174 Mitchell, M., Muftakhidinov, B., Winchen, T., Wilms, A., Schaik, B. van, badshah400,
175 Mo-Gul, Badger, T. G., Jędrzejewski-Szmek, Z., kensington, & kylesower. (2020).
176 *Markumitchell/engauge-digitizer: nonrelease* (Version v12.2.1). Zenodo. <https://doi.org/10.5281/zenodo.3941227>
- 177
- 178 OriginLab Corporation. (2026). *OriginPro 2026*. <https://www.originlab.com>
- 179 Pallets Projects. (2026). *Click: Composable command line interface toolkit* (Version 8.3.3).
180 <https://click.palletsprojects.com/>
- 181 Rohatgi, A. (2024). *WebPlotDigitizer* (Version 5.2). <https://automeris.io>
- 182 The echemdb development team. (2021). *Electrochemistry-data*. GitHub repository. <https://github.com/echemdb/electrochemistry-data/tree/0.8.4>
- 183
- 184 Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., Reddy, T., Cournapeau, D.,
185 Burovski, E., Peterson, P., Weckesser, W., Bright, J., Walt, S. J. van der, Brett, M.,
186 Wilson, J., Millman, K. J., Mayorov, N., Nelson, A. R. J., Jones, E., Kern, R., Larson, E.,
187 ... Vázquez-Baeza, Y. (2020). SciPy 1.0: Fundamental algorithms for scientific computing
188 in python. *Nature Methods*, 17(3), 261–272. <https://doi.org/10.1038/s41592-019-0686-2>
- 189 Wilkinson, M. D., Dumontier, M., Aalbersberg, Ij. J., Appleton, G., Axton, M., Baak, A.,
190 Blomberg, N., Boiten, J.-W., Silva Santos, L. B. da, Bourne, P. E., Bouwman, J., Brookes,
191 A. J., Clark, T., Crosas, M., Dillo, I., Dumon, O., Edmunds, S., Evelo, C. T., Finkers,
192 R., ... Mons, B. (2016). The FAIR guiding principles for scientific data management and
193 stewardship. *Scientific Data*, 3(1). <https://doi.org/10.1038/sdata.2016.18>